SyncleRx

Executive Summary

Client
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Background

SyncleRx is an early-stage healthcare technology initiative transforming medication synchronization (Med Sync) for independent community pharmacies. The initial MVP of the technology was built by a small team of Carnegie Mellon Information Systems students and is led by two Pittsburgh-native founders with 40+ years of combined experience in independent pharmacy and product strategy. The SyncleRx platform analyzes pharmacy management system data in real-time to identify synchronization gaps and deliver explainable, actionable recommendations. By streamlining workflows and reducing cognitive burden, SyncleRx helps pharmacists scale patient management, reduce human error, and optimize their time, thereby enabling them to dedicate more time to care for the patients and communities they serve.

Project Description

Project Opportunity

Medication synchronization (Med Sync) today is often managed through manual processes that require significant staff time and are prone to errors. Pharmacists must account for complex factors such as varying day supplies, late pickups, insurance constraints, and new prescriptions. The lack of standardized tools leads to inefficiencies and missed opportunities for proactive patient care. This project aimed to identify how a structured, technology-driven approach could improve consistency, reduce workload, and enhance decision-making in med sync workflows.

Project Vision

Our vision was to design a lightweight, intuitive system that supports pharmacy staff in identifying and resolving synchronization issues systematically. Rather than replacing existing systems, the goal was to demonstrate how structured logic and clear recommendations could improve workflow efficiency and decision consistency. The system would surface key issues—such as non-30-day supplies or misaligned refill dates—and guide users toward actionable next steps.

Project Outcomes

We developed a standalone web-based prototype that models med sync workflows and generates actionable recommendations. The system identifies key cases such as first-time synchronization, late pickups requiring resynchronization, insufficient remaining quantities, and new prescriptions for already-synced patients.

From a **process perspective**, we translated informal pharmacy decision-making into a structured framework that can be consistently applied, reducing manual work. From a **technology perspective**, we built a React-based frontend with backend logic to process prescription data and generate flags and recommendations. From a **people perspective**, we worked closely with the client to validate assumptions and refine logic to better reflect real-world pharmacy workflows.

While the prototype is not integrated into existing pharmacy systems, it demonstrates the feasibility and value of a structured med sync approach.

Project Deliverables

The final deliverables include a fully functional web-based prototype, structured datasets for testing different synchronization scenarios, and documentation detailing system logic, assumptions, and design decisions. We also provided clear descriptions of the flagging system and recommendation engine, along with example workflows that demonstrate how the system can be used in practice.

Recommendations

We recommend that SyncleRx focus on expanding system coverage and improving workflow integration to maximize long-term impact and adoption. Specifically, extending the system to support additional Med Sync edge cases would further reduce reliance on manual intervention and increase the proportion of patients fully supported by system recommendations. In addition, enabling direct CSV exports from DRX that align with the system's required schema would eliminate the current manual data preparation step, reducing friction and making the workflow more scalable in daily use. Finally, exploring integration pathways with DRX or similar pharmacy management systems would allow for real-time data access and deeper embedding into existing workflows, significantly increasing efficiency and consistency. The impact of these improvements can be measured through reductions in daily review time, increased system usage, and improved consistency in synchronization decision-making.

Student Consulting Team

Autumn Qiu served as project manager and software developer. She is a third-year student majoring in Information Systems with minors in SWE and Product Management. She will be interning at Mastercard this summer and is looking toward a career as a full-stack SWE.

Evelyn Zheng served as a project contributor and software developer. She is a fourth-year student majoring in Information Systems and is interested in building user-centered technology solutions as a full-stack software engineer.

Sean Jiang served as a project contributor and software developer. He is a third-year student studying Information Systems and Artificial Intelligence. He is passionate about building intelligent software that connects AI with real-world applications.